Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/29/2025
PI Name:	Bowles, Dawn Ph.D.		
Project Title:	Awakening Endogenous Retroviruses with the Space Environment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27710-0001	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	10/01/2019	**End Date:**	12/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2022 per L. Barnes-Moten/JSC (Ed., 6/20/23) NOTE: End date changed to 06/30/2021 per L. Barnes-Moten/JSC (Ed., 4/15/21) NOTE: End date changed to 3/31/2021 per NSSC information (Ed., 11/9/20)		
Key Personnel Changes/Previous PI:			
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Task Description:	Approximately 8% of the human genome is composed of Endogenous Retrovirus (ERVs). These genetic elements have helped shape humans as they exist today. ERVs are normally maintained in a silenced state but non-specific activation of ERVs can occur through application of exogenous stressors, and may have detrimental consequences to the host. Indeed, within the last 5 years a number of studies have demonstrated ERVs to be associated with autoimmune diseases, cancer, and neurological diseases. Extended space travel will expose astronauts to the space radiation and microgravity environments; both of these stressors may influence genomic modifications that may result in non-specific activation of ERVs. Activation of ERVs may result in alterations to molecular pathways within different cell types that might influence negative pathogenic outcomes during space flight. The key central objective of this proposal is to understand how the physical space environment might influence activation of ERVs. We will accomplish this research objective in two aims. In the first aim we will utilize a bioreactor to grow cells in a microgravity environment and evaluate the cells by molecular and immunofluorescence techniques for evidence of ERV activation under this space stressor. The second aim will utilize an established tissue repository that contains multiple tissues from mice exposed to various types and doses of space radiation. We will examine these tissues molecularly and histologically for evidence of ERV activation. This project is significant in that identification of ERVs that respond to specific space conditions may function as early, surrogate markers of putative genomic change. Furthermore, expression of ERV-encoded proteins may be seen as foreign and elicit autoimmune responses.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Overall some progress has been made demonstrating the increase in cells with dsRNA (and possibly endogenous retrovirus reactivation) following both gamma and galactic cosmic radiation (GCR) exposure. This project is significant in that identification of ERVs that respond to specific space conditions may function as early, surrogate markers of putative genomic change. Furthermore, expression of ERV-encoded proteins may be seen as foreign and elicit autoimmune responses.
Task Progress:	In this study, we developed methods to detect endogenous retrovirus activation focusing on the detection of dsRNA.To develop our dsRNA assays, we used the J2 anti-dsRNA IgG2a monoclonal antibody. These antibodies have become the gold standard in dsRNA detection. J2 can be used to detect dsRNA intermediates of viruses as diverse as Hepatitis C virus, Dengue virus, rhinovirus, Chikungunya virus, Rabies virus, Poliovirus, and many more in cultured cells and in fixed paraffin-embedded histological samples. Flow cytometry experiments using J2 antibody were performed in A549 cells exposed to gamma, galactic cosmic radiation (GCR), and gamma+ microgravity. Notably, increases in average median fluorescence intensity (aMFI) were seen in cells exposed to 8 Gy gamma compared to unirradiated control cells (1.30-fold increase/p=0.0175) at 2 days post radiation. Combining two stressors (10 Gy Gamma and microgravity) showed a 1.73-fold increase (p=0.0002) in aMFI compared to normal gravity controls and a 1.94-fold increase (p<0.0001) in aMFI compared to 10 Gy Gamma only treated samples. Five-day post-radiation results, however, showed statistical significance across all experimental treatment groups; 8 Gy Gamma, 10 Gy Gamma, and microgravity-treated samples showed a 1.33-fold increase (p=0.0155), 2.13-fold increase (p=0.0004), and 2.41-fold increase (p=0.0004) in aMFI, respectively. Combined effects of 10 Gy Gamma and microgravity at the 5-day post-radiation time point showed a 4.72-fold (p<0.0001) increase in aMFI compared to normal gravity controls and a 1.96-fold (p=0.0215) increase in aMFI compared to microgravity only treated samples. One-day post-radiation samples irradiated with GCR5-ion displayed a 1.21-fold increase in aMFI although it was not statistically significant (p=0.47). We also utilized the J2 antibody to evaluate dsRNA in tissues from mice (lung, liver, heart, brain) from mice exposed to gamma, oxygen ion, iron ion, or GCR radiation compared to age-matched unirradiated mice. Statistically, the tissues most affected were: 1) livers in iron-irradiated mice; 2) brains in oxygen-irradiated mice; and 3) livers and brains of gamma-irradiated mice. The number of animals per radiation group examined was relatively low (n=3), so we pooled the average intensities from iron, oxygen, and gamma-irradiated mice. The intensities measured from the livers, brains, lungs, and hearts were all greater than the corresponding tissues from the unirradiated control animals. The intensities measured in the livers and hearts met statistical significance, but the intensities measured in the brains and lungs failed to meet statistical significance. Overall, an increase in the presence of double-stranded RNA as evidenced by the increase in immunofluorescence intensity of J2 staining in radiation conditions versus control was seen in both acute (cell-based studies) as well as in chronic (mouse-tissue based) radiation studies. Notably, the tissues evaluated in this study were obtained from mice that had been exposed to various types of radiation 17-18 months prior to euthanasia and tissue banking. A single dose of radiation has a long-lasting effect on the presence of double-stranded DNA indicating a continuing pro-inflammatory environment following the initial exposure. Future work would consist of evaluating tissues obtained from animals euthanized at earlier time points post-radiation to better understand the kinetics of dsRNA expression post-radiation.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Pla MM, Lee FH, Smith MF, Lobo AA, Diaz TE, Cockrell AS, Bowles DE. "Double-stranded RNA and reverse transcriptase signatures of endogenous retroviruses after exposure to spaceflight environment related stressors." Life Sci Space Res. 2026 Jun 19. Online ahead of print. https://doi.org/10.1016/j.lssr.2026.06.005 , Jun-2026